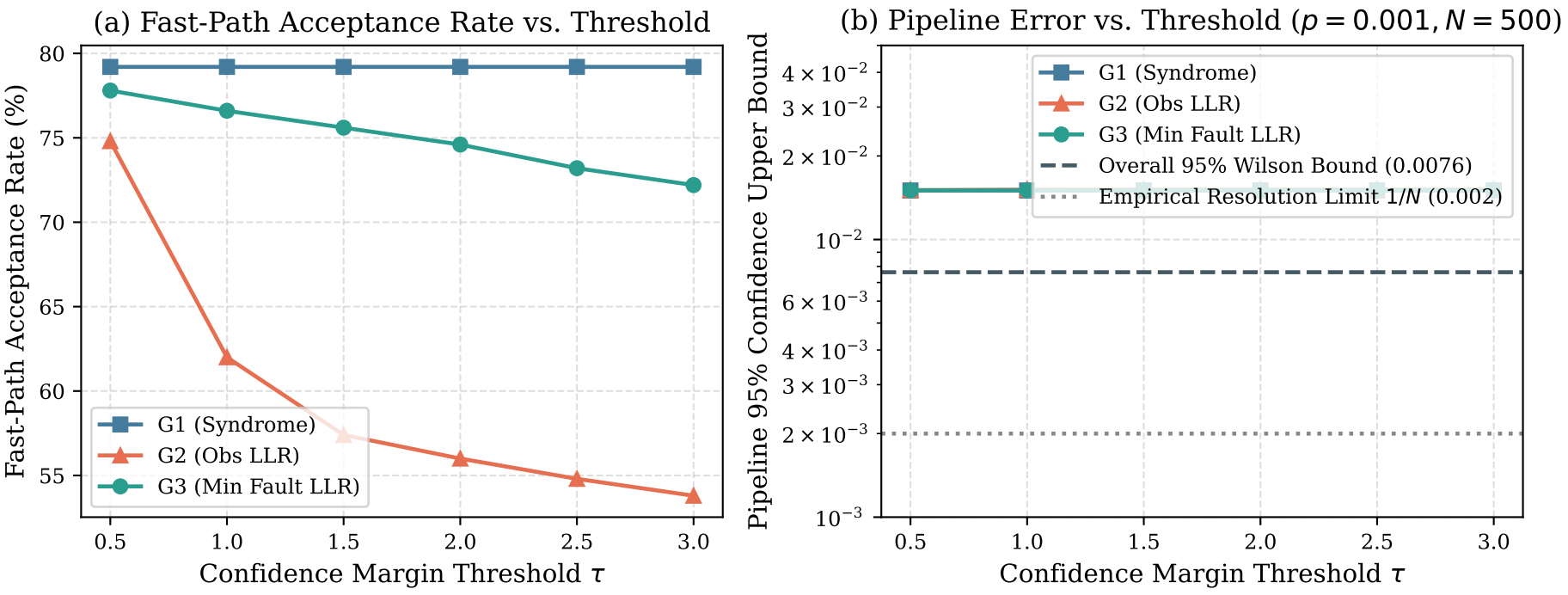
(a) Fast-Path Acceptance Rate vs. Threshold
Fast-Path Acceptance Rate (%)
80
75
70
65
60
55
0.5
1.0
1.5
2.0
2.5
3.0
Confidence Margin Threshold τ
G1 (Syndrome)
G2 (Obs LLR)
G3 (Min Fault LLR)
(b) Pipeline Error vs. Threshold ($p = 0.001, N = 500$)
Pipeline 95% Confidence Upper Bound
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
10^{-3}
0.5
1.0
1.5
2.0
2.5
3.0
Confidence Margin Threshold τ
G1 (Syndrome)
G2 (Obs LLR)
G3 (Min Fault LLR)
Overall 95% Wilson Bound (0.0076)
Empirical Resolution Limit $1/N$ (0.002)